



Safe Artificial Intelligence

Project Announcement

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Project Announcement

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- **Final Project and Project Proposal**

- **Project proposal (5/12, 5/14)**

- Prior to the final project, students are expected to apply the knowledge and coding skills acquired during the course to real-world datasets.

- **Final project (6/16)**

- Investigating safety alignment and exploring potential solutions in AI models using a dataset.
 - The project should focus on one of the following main themes (as covered in class):
 1. **Robustness** (e.g., find a new type of failures or improve robustness of AI models)
 2. **Explainability** (e.g., extract a new explanation on the datasets)
 3. **Privacy** (e.g., find a new type of privacy leakage in AI models or remove private info than existing methods)
 - Each team should submit 4-page short paper.
 - We will use overleaf (later we cover this).

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- Project proposal

- Objective: **Introduce YOUR TEAM's task, dataset, and models to other students!**
 1. **Dataset:** Specify the dataset you will use.
 - Clearly define the task.
 - Explain the dataset, the number of features, and describe the features.
 2. **Model:** Describe the model you plan to use or develop.
 - You may use models such as the Multilayer Perceptron (MLP) or Convolutional Neural Networks (CNN) covered in class.
 - **You may employ other models**; however, **if you choose to do so, you must briefly explain the model architecture and why you choose.**
 3. **Performance:** Clearly state the specific objectives and results on the dataset.
 - Investigate and report **the current state-of-the-art performance** for your task, referencing benchmarks (**e.g., Papers with Code, Github, or recent research papers**).
 - Summarize the results for your own model, training recipe, and performance.
 4. **What you gonna do?**
 - Explain briefly what you gonna do in the final project.
 - **No evaluation on this. Keep it simple.**

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- **Project proposal**

- **Presentation**

- All presentation materials must be in English, but you may present in any language you prefer.
 - Each presentation should be 15-20 minutes long.
 - Be prepared for a Q&A session following your presentation.

- **Submit the following materials to “Project Proposal” board:**

- 1. **Presentation materials.**

- 2. **Files** that can be shared to other students with

- Used models
 - Used datasets
 - Training recipes and results /w explanations (**Github link recommended**)

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- **Project proposal**
 - **Evaluation Criteria (evaluated by Professor)**
 - **Creativity (30%):**
 - Does the team offer an unique dataset?
 - Point = $60 / \max(\text{\#teams selected the same dataset}, 2)$
 - **Analysis (40%):**
 - The depth and quality of the investigation.
 - Has the team critically analyzed datasets, tasks, training recipes of models, performance of each model?
 - How closely did the achieved performance approach the state-of-the-art (SOTA) performance?
 - **Delivery (30%):**
 - The effectiveness of the presentation such as NOT reading a script.
 - Does the team maintain audience engagement and handle the Q&A session.

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- **Tips for Project Proposal**

1. Please avoid selecting large-scale datasets, such as ImageNet. Without access to high-end GPUs, it is not feasible to train deep learning models on such datasets using platforms like Google Colab.
2. Detail explanation on the task, dataset, and models will be important.
3. Start working on the project early. As you progress, you may find it necessary to change the dataset or adjust your approach based on the analysis.

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- **After Project Proposal?**

- All teams can choose one of datasets presented by other teams (or their own program or other).
 - Or you can also find a new dataset.
- Once the dataset is finalized, **in the final project**, students are expected to conduct an analysis of both the dataset and the models, focusing on the one of the following aspects:
 1. **Robustness**
 2. **Explainability**
 3. **Privacy**
- Details will be announced later.

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- **Special Rule**

- <https://judgementday.aim-intelligence.com/arena>
 - Select a scenario to begin your adversarial attack.
 - Each scenario presents an AI agent operating in a high-stakes environment.
 - Submit crafted inputs to trigger or suppress specific model behaviors.
- When you join this competition, no reduction on the creativity.
- However, it is NOT easy. You should study
 - Transformer to LLM
 - AI Agent